[D] Traits

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Traits

Allele

Aa

Monogenic trait

actual genetic material

expression of genes

☆ Genotype vs Phenotype

Possible combinations of alleles (Aa, AA, aa)

Possible observable traits [(Aa, AA) or (aa)]

☆ Homozygous vs Heterozygous

Both alleles same — AA or aa

Different alleles — Aa

☆ Dominant vs Recessive (alleles)

A — a

- Chromosomes are associated w/ passing of traits

Mendel's Principles

Law of Segregation: Since offspring inherit one allele from each parent, traits cannot be blended - either inherits from one parent or the other

Law of Independent Assortment: genes are inherited separately from one another

Sex-Linked Traits

- controlled by genes on X or Y chromosomes
- Y chromosome has male traits, but X chromosome has many other functions
- X, Y <u>not</u> homologous (no match)
  - more likely to produce disorders in males (e.g. colorblindness, hemophilia)
    - colorblindness: color sensitive rhodopsin proteins
    - hemophilia: blood clots
  - females can be carriers (heterozygous)

father has hemophilia: XY   XX

XY or YX   XX or XX

Non-Mendelian Inheritance

Non-Mendelian Inheritance

XY or YX   XX or XX
son = 50%   daughter = 50%

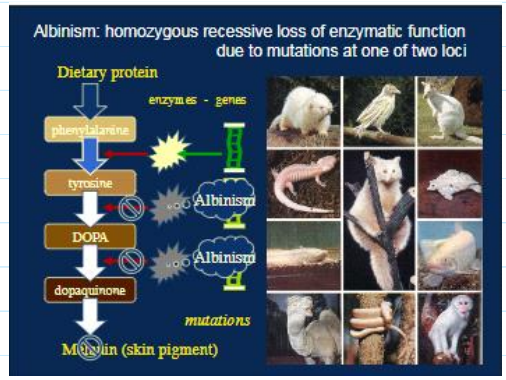
Mother: XY XX
YX YX son: 100%
XX XX daughter: carrier

- Polygenic/continuous traits
- coded by multiple alleles
- examples: blood type, skin color
- Phenotypes can also be affected by environment (e.g. height)

Mutation

- new changes in DNA that cause genetic variation
- occurs when codon is changed, altering the amino acid produced
- mechanical errors during gene replication

Point Mutation: affects one part of a protein.
⇒ example: Albinism occurs when there is a mutation in one locus in the production of melanin



Pedigree Diagrams